

Executive Summary – Hotel Booking Cancellation Analysis

1. Project Overview

The Hotel Booking Cancellation Analysis project analyzes hotel booking patterns, cancellations, guest types, room preferences, hotel categories, and monthly booking trends. The dashboard provides an interactive view of booking performance for City Hotels and Resort Hotels across the years 2015–2017.

The objective is to understand the major patterns behind booking cancellations and identify customer segments and booking characteristics that may require greater attention.

2. Key Performance Indicators

- **Total Bookings:** 119,390
- **Total Cancellations:** 44,224
- **Overall Cancellation Rate:** Approximately **37.0%**
- **Analysis Period:** 2015–2017

This means that approximately 37 out of every 100 bookings were cancelled in the analyzed dataset.

3. Hotel Type and Cancellations

The dashboard shows a clear difference between City Hotels and Resort Hotels.

- **City Hotel:** 79,330 total bookings and 33,102 cancellations, giving a cancellation rate of approximately 41.7%.
- **Resort Hotel:** 40,060 total bookings and 11,122 cancellations, giving a cancellation rate of approximately 27.8%.

City Hotels therefore account for the larger share of both total bookings and cancellations in the dataset.

4. Room Type and Cancellations

The dashboard compares Desired and Un-Desired room types:

- **Desired rooms:** 104,473 bookings and 43,422 cancellations, approximately 41.6%.

- **Un-Desired rooms:** 14,917 bookings and 802 cancellations, approximately 5.4%. The majority of bookings are for desired room types, and cancellations are also concentrated within this category.

5. Guest Type Analysis

The dashboard compares Couples, Families, and Singles.

- **Couples:** 81,560 bookings and 32,424 cancellations — approximately 39.8%.
- **Families:** 15,253 bookings and 5,245 cancellations — approximately 34.4%.
- **Singles:** 22,577 bookings and 6,555 cancellations — approximately 29.0%.

Couples represent the largest booking segment and also have the highest cancellation rate among the three guest groups shown.

6. Monthly Booking Trends

The monthly visualization shows that booking volumes and cancellations vary throughout the year.

Booking activity increases during several months, with relatively high booking volumes visible around April through October, while July and August show particularly high booking activity in the displayed dashboard.

The monthly pattern suggests that hotel demand and cancellations may be influenced by seasonality, holidays, travel periods, and customer booking behavior.

7. Overall Business Insights

The dashboard highlights several important findings:

1. The dataset contains 119,390 bookings, of which 44,224 were cancelled.
2. The overall cancellation rate is approximately 37.0%.
3. City Hotels have a higher cancellation rate (41.7%) than Resort Hotels (27.8%).
4. Desired room types account for the majority of bookings and cancellations.
5. Couples represent the largest guest segment and have the highest cancellation rate among the displayed guest types.
6. Singles have the lowest cancellation rate among the three guest categories.
7. Booking and cancellation volumes vary across months, indicating seasonal patterns in hotel demand.
8. The dashboard's year filter allows the booking patterns to be examined across 2015, 2016, and 2017.

Recommendations

1. Focus on City Hotel Cancellations

Since City Hotels show a higher cancellation rate, management can investigate factors such as booking lead time, deposit requirements, pricing, and customer booking behavior specifically for City Hotels.

2. Strengthen Cancellation Management

Introduce suitable cancellation policies, reminders, and flexible rebooking options to reduce avoidable cancellations while maintaining customer satisfaction.

3. Focus on Couple Bookings

Couples account for the largest guest segment and show a relatively high cancellation rate. Personalized offers, flexible booking options, and pre-arrival engagement could be considered for this segment.

4. Use Seasonal Planning

The monthly trends can be used to plan staffing, room availability, pricing, and promotional campaigns around periods of higher booking demand.

5. Analyze Desired Room Bookings

Since most bookings and cancellations are associated with desired room types, management should examine whether factors such as room availability, price changes, booking lead time, or room allocation are contributing to cancellations.

6. Monitor Cancellation Trends by Year

The year slicer can be used to compare 2015, 2016, and 2017 and identify whether cancellation patterns are changing over time.

Conclusion

The Hotel Booking Cancellation Analysis dashboard provides a comprehensive view of booking volume and cancellation behavior across hotel types, room preferences, guest categories, and months. The overall cancellation rate of approximately 37.0% highlights the importance of understanding cancellation behavior.

The analysis shows that City Hotels, desired room bookings, and couple bookings represent important areas for further investigation because of their booking volumes and cancellation patterns. Combining these insights with seasonal analysis and year-wise comparisons can help hotel management improve booking policies, customer engagement, and operational planning.